Problem Solving

Problem solving - Intro

- Computerization - Indicate the use of computer to develop software in order to automate any routine human task efficiently.
- Computers themselves cannot solve a problem.
- Precise step-by-step instructions should be given by us to solve the problem
- The success of a computer in solving a problem depends on how correctly and precisely we define the problem, design a solution (algorithm) and implement the solution (program) using a programming language.

Problem solving - Intro

- When problems are straightforward and easy, the solution is easy to find.
- A complex problem requires a methodical approach to find the right solution
- Problem solving begins with the precise identification of the problem and ends with a complete working solution in terms of a program or software.

Steps in Problem solving

1. Identify the problem
2. Understand the problem
3. Identify alternate ways to solve the problem
4. Select the best way to solve the problem from the list of alternative solutions
5. List instructions that enable you to solve the problem using the selected solution
6. Evaluate the solution

- **Algorithmic solutions**

Problems that can be solved with a series of actions, solutions by steps, eg. Baking a cake

- **Heuristic solutions**

Solutions are not straightforward

Require reasoning built on knowledge and experience and a process of trial and error

Eg. Buying the best stock, whether to expand a company….

Field of computers that deal with heuristic types of problems is called artificial intelligence

Problem solving in computers

1. Understanding the problem: Here we try to understand the problem to be solved in totally. Before with the next stage or step, we should be absolutely sure about the objectives of the given problem.

2. Analyzing the problem: After understanding thoroughly the problem to be solved, we look different ways of solving the problem and evaluate each of these methods. The idea here is to search an appropriate solution to the problem under consideration. The end result of this stage is a broad overview of the sequence of operations that are to be carried out to solve the given problem

Problem solving in computers contd…

3. Developing the solution: Here the overview of the sequence of operations that was the result of analysis stage is expanded to form a detailed step by step solution to the problem under consideration.

4. Coding and implementation: The last stage of the problem solving is the conversion of the detailed sequence of operations in to a language that the computer can understand. Here each step is converted to its equivalent instruction or instructions in the computer language that has been chosen for the implantation.

Algorithm

Definition - *A set of sequential steps usually written in Ordinary Language to solve a given problem is called* ***Algorithm***.

An algorithm can be defined as **“a complete, unambiguous, finite number of logical steps for solving a specific problem “**

- One problem, many solutions – many algorithms
- Selection depends on various factors like reliability, accuracy and easy to modify
- The most important factor in the choice of algorithm is the time requirement to execute it, after writing code in High-level language with the help of a computer.
- The algorithm which will need the least time when executed is considered the best.

Steps involved in algorithm development

- **Step1. Identification of input**: For an algorithm, there are quantities to

be supplied called input and these are fed externally. The input is to be identified first for any specified problem.

- **Step2: Identification of output**: From an algorithm, at least one quantity

is produced, called for any specified problem.

- **Step3 : Identification the processing operations** : All the calculations

to be performed in order to lead to output from the input are to be identified in an orderly manner.

Steps involved in algorithm development…

- **Step4 : Processing Definiteness** : The instructions composing the algorithm must be clear and there should not be any ambiguity in them.
- **Step5 : Processing Finiteness** : If we go through the algorithm, then for all cases, the algorithm should terminate after a finite number of steps.
- **Step6 : Possessing Effectiveness :** The instructions in the algorithm must be sufficiently basic and in practice they can be carried out easily.

Properties

1. Finiteness: An algorithm must terminate in a finite number of steps

2. Definiteness: Each step of the algorithm must be precisely and unambiguously stated

3. Effectiveness: Each step must be effective, in the sense that it should be primitive easily convert able into program statement) can be performed exactly in a finite amount of time.

4. Generality: The algorithm must be complete in itself so that it can be used to solve problems of a specific type for any input data.

5. Input/output: Each algorithm must take zero, one or more quantities as input data produce one or more output values. An algorithm can be written in English like sentences or in any standard representation sometimes, algorithm written in English like languages are called Pseudo Code

Example: Find the average of three numbers

Step 1 : Read the numbers a, b, c

Step 2 : Compute the sum of a, b and c

Step 3 : Divide the sum by 3

Step 4 : Store the result in variable d

Step 5 : Print the value of d

Step 6 : End of the program

Example: To calculate the simple interest

- Simple interest = P*N* R/100. Where P is principle Amount, N is the number of years and R is the rate of interest.

Step 1: Read the three input quantities' P, N and R.

Step 2 : Calculate simple interest as

Simple interest = P* N* R/100

Step 3: Print simple interest.

Step 4: Stop.

Example: Area of Triangle

Let b, c be the sides of the triangle ABC and A the included angle between the given sides.

Step 1: Input the given elements of the triangle namely sides b, c and angle between the sides A.

Step 2: Area = (1/2) *b*C* sin A

Step 3: Output the Area

Step 4: Stop.

Example: To find the largest of three numbers X, Y, Z.

Step 1: Read the numbers X,Y,Z.

Step 2: if (X > Y)

Big = X

else BIG = Y

Step 3 : if (BIG < Z)

Step 4: Big = Z

Step 5: Print the largest number i.e. Big

Step 6: Stop.

While writing an algorithm, it is required to clearly identify the following:

- The input to be taken from the user
- Processing or computation to be performed to get the desired result
- The output desired by the user

Representation of Algorithms

Two methods

1) Flowchart
2) Pseudo-code

Either of the methods can be used to represent an algorithm while keeping in mind the following:

• It showcases the logic of the problem solution, excluding any implementation details

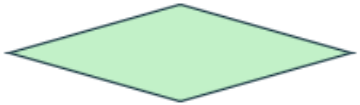
• It clearly reveals the flow of control during execution of the program

Flowchart

- **A Visual Representation of Algorithms**
- A diagram made up of boxes, diamonds and other shapes, connected by arrows.
- Each shape represents a step of the solution process and the arrow represents the order or link among the steps.

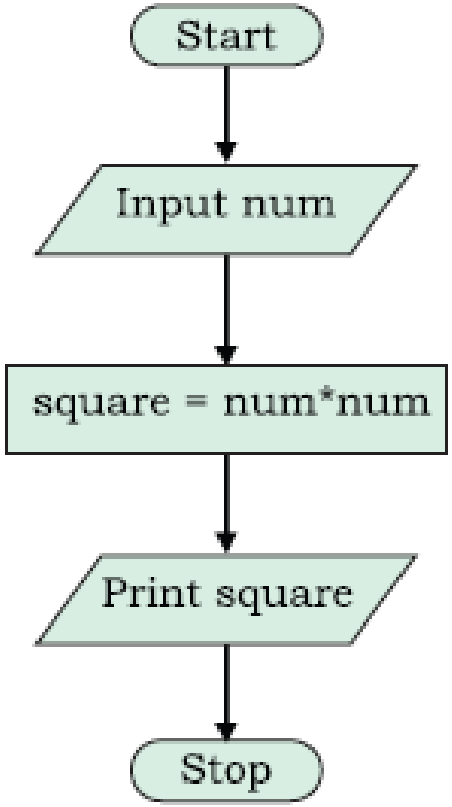
There are standardized symbols to draw flowcharts.

Shapes or symbols in flow charts

Symbol	Name	Function
	Oval	Represents the start or end of a process
	Rectangle	Denotes a process or operation step
	Arrow	Indicates the flow between steps
	Diamond	Signifies a point requiring a yes/no
	Parallelogram	Used for input or output operations

Write an algorithm to find the square of a number.

- Step 1: Input a number and store it to num
- Step 2: Compute num * num and store it in square
- Step 3: Print square



To solve the problem of a non-functioning light bulb

